# Alert System Using Reinforcement Learning (PPO)

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## Problem Statement

Challenges at Road Intersections

- Many rural intersections do not have traffic signals or automated control systems.
- Vehicles approach from multiple directions with limited coordination.
- Poor visibility and lack of structured traffic management increase safety risks.

## Key Issues

- High risk of vehicle conflicts and collisions
- Unpredictable vehicle behavior at intersections
- Inefficient pacing when multiple vehicles arrive simultaneously
- Delays and confusion due to absence of traffic control

An adaptive system is required to:

- Provide real-time alerts to drivers or teleoperators
- Enable safe decision-making at signal-free intersections
- Maintain smooth and efficient traffic movement

# Why Reinforcement Learning Approach

## Limitation of heuristic-based approaches

- Require manual rule design for multiple variables such as speed, distance, pose etc.
- Hard to manage complex interactions between vehicles approaching from different directions
- Rules become difficult to scale as dataset variables increase (vehicle type, network delay, intersection type, etc.)
- Cannot easily adapt to dynamic multi-vehicle scenarios at intersections

## Reinforcement Learning Approach

- RL learns data-driven alert policies from simulation by adapting to dynamic traffic
- Optimizes long-term safety and traffic efficiency for multi-vehicle system
- Handles multiple interacting variables such as speed, distance, direction etc.
- Easier to generalize to different intersection types

# Alert System: Dataset preparation

| Case | Direction | Speed (MPH) | Distance (meter) | Pose (degree) | Latency (ms) | Vehicle Type |
|---|---|---|---|---|---|---|
| 3 way | E | 45 | 62 | 110 | 78 | Van |
| | N | 62 | 58 | 120 | 92 | Car |
| | S | 55 | 49 | 95 | 80 | Car |
| 4 way | E | 60 | 52 | 150 | 108 | Truck |
| | W | 55 | 85 | 80 | 86 | Motor Bike |
| | N | 48 | 74 | 120 | 180 | Car |
| | S | 68 | 80 | 55 | 132 | Van |

# Alert System: Dataset preparation

| Case | Dir | Speed (MPH) | Distance (meter) | Heading angle | Vehicle Type |
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## **Alert System :** RL model Reward Function

Total reward is composed of four components:

1. Safety component
2. Efficiency component
3. Priority component
4. Network component

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# Alert System : RL model Reward Function

$$R_{\text{total}} = \sum_{i=1}^{N} \Bigg[ w_s - \alpha \cdot \text{collision}_i - \beta \sum_{j \neq i} C_{i,j}\, H_{i,j}\, W_{\text{type}} \exp\left( -\frac{TTC_{i,j}}{\tau} \right)$$

$$+ w_e \left( \gamma \frac{v_i}{v_{\text{desired},i}} - \delta \frac{\text{waiting}_i}{T_{\max}} \right)$$

$$+ w_p \mu \left( \lambda_1 \frac{\text{waiting}_i}{T_{\max}} + \lambda_2 \frac{1}{\text{distance}_i + 1} + \lambda_3 \frac{v_i}{v_{\text{desired},i}} + \lambda_4\, \text{conflict density}_i \right) \Bigg]$$

# **Alert System :** RL model Reward Function

Simplified form,

$$R_{\text{total}} = \sum_{i=1}^{N} \left( w_s R_i^{safety} + w_e R_i^{efficiency} + w_p R_i^{priority} \right)$$

# Alert System : RL model Reward Function

## Safety Term

$$R_i^{safety} = -\alpha \cdot \text{collision}_i - \beta \sum_{j \neq i} C_{i,j} H_{i,j} W_{\text{type}} \exp\left(-\frac{TTC_{i,j}}{\tau}\right)$$

## Efficiency Term

$$R_i^{efficiency} = \gamma \frac{v_i}{v_{\text{desired},i}} - \delta \frac{\text{waiting}_i}{T_{\max}}$$

## Priority Term

$$R_i^{priority} = \mu \left( \lambda_1 \frac{\text{waiting}_i}{T_{\max}} + \lambda_2 \frac{1}{\text{distance}_i + 1} + \lambda_3 \frac{v_i}{v_{\text{desired},i}} + \lambda_4 \text{conflict density}_i \right)$$

# Alert System : RL model Reward Function

## Rewards (positive)

### (a) Maintaining Desired Speed

$$\gamma \frac{v_i}{v_{\text{desired},i}}$$

**Meaning**

- Encourages vehicles to move close to their desired speed.
- Promotes smooth traffic flow and prevents unnecessary slowing.

**Effect**

✔ Higher reward when

$v_i \approx v_{desired,i}$

# Alert System : RL model Reward Function

## Rewards (positive)

### (b) Priority-based Waiting Compensation

$$\lambda_1 \frac{waiting_i}{T_{max}}$$

**Meaning**

- Gives priority to vehicles that have waited longer.
- Ensures fairness at the intersection.

**Effect**

✔ Reward increases as waiting time increases.

# Alert System : RL model Reward Function

## Rewards (positive)

### (c) Distance-based Urgency

$$\lambda_2 \frac{1}{distance_i + 1}$$

**Meaning**

- Vehicles closer to the intersection get higher priority.

**Effect**

✔ Reward increases as vehicle approaches the conflict point.

# Alert System : RL model Reward Function

## Rewards (negative)

### (a) Collision Penalty

$$-\alpha \cdot collision_i$$

**Meaning**

- Strong penalty when a collision occurs.

**Effect**

❌ Heavy negative reward.

This is typically the **largest penalty in the function**.

# Alert System : RL model Reward Function

## Rewards (negative)

### (b) Time-To-Collision Risk Penalty

$$-\beta \sum_{j \neq i} C_{i,j} H_{i,j} W_{type} \exp\left(-\frac{TTC_{i,j}}{\tau}\right)$$

**Meaning**

- Penalizes situations where vehicles approach collision risk.
- TTC penalty increases when vehicles are close to colliding.

**Effect**

❌ Strong penalty when TTC is small.

Encourages **proactive collision avoidance**.

# Alert System : RL model Reward Function

**Rewards (negative)**

### (c) Waiting Time Penalty (Efficiency)

$$-\delta \frac{waiting_i}{T_{max}}$$

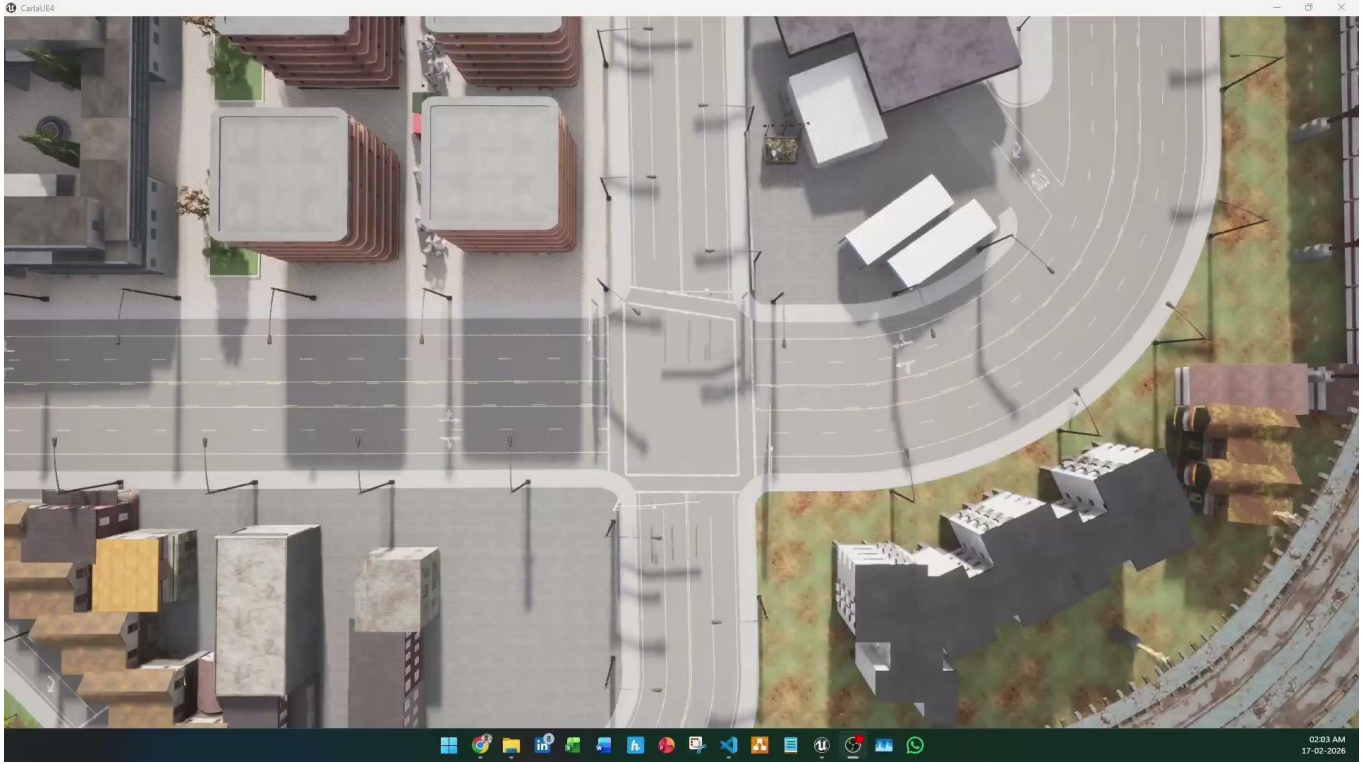
**Meaning**

- Discourages vehicles from waiting too long.

**Effect**

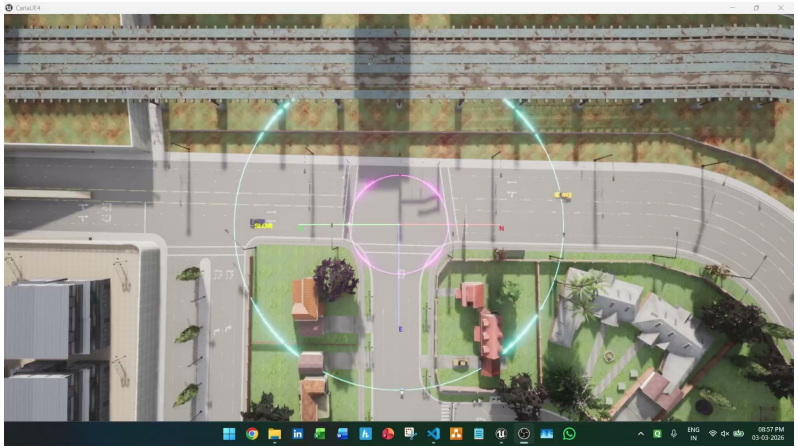
❌ Penalizes traffic delay.

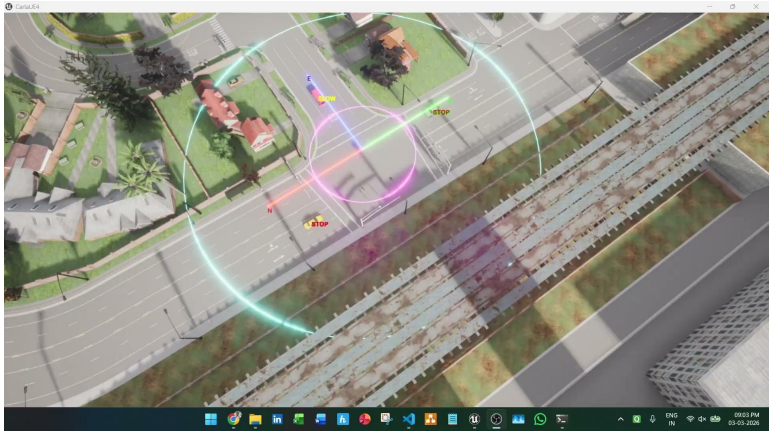
# No Alert System



# Alert System : RL model Test

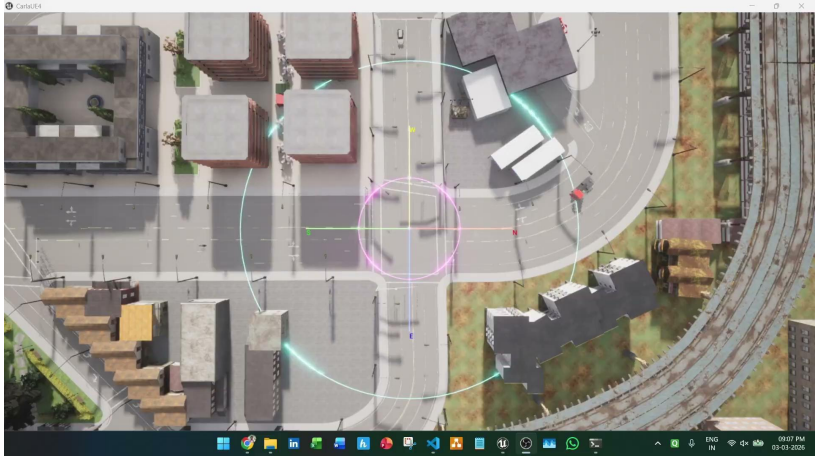
## 3 Way Intersection





# Alert System : RL model Test

## 4 Way Intersection



# Future Steps

- Formulate network component into the reward function.
- Improve distance-based urgency in Carla simulation for better real world results.
- We are going to use hybrid approach (offline +online) for our formulation.
- Plan to publish the work in the future.